Novelty Assessment:
AutoPos — 3D Anchor Self-Calibration from Inter-Anchor Ranging Alone

System-Paper Positioning and Prior-Art Delineation

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This note is the novelty assessment for the AutoPos concept (Paper A, a metrology/deployment systems paper). It is one half of a deliberately split, non-overlapping two-paper program. Paper A reports the working system, its ground-truth-validated accuracy, and the vertical limit it does not cross; the companion mechanism paper (Paper B) analyzes the delay–layout coupling behind that limit. The delay-estimation result appears in Paper A only as an empirical system effect ("modelling per-device delay helps"); Paper B characterizes the coupling itself — its headline *results are an anchor-side common-delay↔layout-scale degeneracy and a Vicon-grounded falsification protocol, while a* tag-side *strand of the same analysis explains why* this *system's residual vertical error is a per-tag delay aliasing rather than a geometry deficit that more geometry could remove.*

1 One-Sentence Thesis

AutoPos is, to our knowledge, the first UWB anchor self-calibration system that recovers full *3D* anchor coordinates jointly with per-device antenna delays from *inter-anchor ranging alone* — with no external sensor (no LiDAR/SLAM/VIO/ultrasound), no mobile survey platform, and no manually surveyed reference positions — and that validates the result against optical ground truth, reporting both the accuracy it reaches and the systematic vertical bias it does not remove. The novelty is a *deployment capability and its honest characterization*, not a new estimator: the individual numerical components are established; what is new is their integration for the inter-anchor-only 3D regime that prior work explicitly avoids, together with ground-truth-free quality diagnostics usable at deployment time.

2 Motivation: the Accuracy Ceiling and the Calibration Bottleneck

The systematic review of 37 UWB+IMU motion-tracking systems by Yogesh et al. (2023) documents that even the *best* reported accuracy is only ∼5 cm: only ∼19% of the systems reach ∼10 cm even under ideal line-of-sight with optically surveyed anchors, the majority report 10–80 cm, and several exceed 1 m — all well short of the ∼1 cm that the review derives from matching optical motion capture, the accuracy "gold standard" it considers clinically relevant. Only *5 of the 37 systems attempt 3D positioning at all*; the other 32 operate purely in 2D, avoiding the vertical axis. The review singles out systematic UWB ranging error as the dominant bottleneck, and its largest component is per-device antenna delay — a constant ranging bias of ∼150 mm per node (Shah et al., 2021; Sang et al., 2019), an order of magnitude above the transceiver's random noise. The follow-up (Yogesh et al., 2024) finds that none of the reviewed systems even apply the manufacturer's recommended delay-calibration procedure — let alone model the delay inside their estimation pipeline — and reports that even *after* that procedure a ∼3–4 cm systematic ranging residual remains.

The systems that do reach high accuracy do so only by importing infrastructure that is unavailable out of the lab: angle-of-arrival base stations and surveyed anchor coordinates (Zihajehzadeh and Park, 2017), or dedicated procedures requiring known reference positions (Kok et al., 2015). This is a *circular dependency*: accurate UWB positioning needs accurately known anchor coordinates, but obtaining those coordinates needs an optical system or surveyed references — precisely what is missing in ambulatory, clinical, and field deployments. Anchor self-calibration is the route out of that circle, and the gap AutoPos targets is that no existing self-calibration method closes it in 3D from inter-anchor ranging alone on validated commodity hardware (Section 3).

Why this system, and why 4+4: the vertical axis has no shortcut. The target application is body-worn motion capture, which sharpens the requirement in a way that drone or robot UWB does not. A hovering platform can resolve its height with an onboard downward range sensor (ultrasound, time-of-flight) or fuse a barometer (Xiang et al., 2025); a body-worn tag cannot — an IMU gives only drifting relative vertical displacement, and a barometer only coarse, drifting absolute height, neither adequate for centimetre-level limb tracking. The *only* drift-free, centimetre-level absolute vertical reference available to a body-worn tag is the UWB geometry itself. The vertical must therefore be carried by pure UWB, which is exactly why the array needs vertical geometric diversity (the balanced 4+4 two layers) rather than four coplanar anchors for the horizontal plus a height sensor on the tag. This is also why what we term the *delay–layout coupling* surfaced here at all: forcing UWB to recover its weakest axis (the vertical) is precisely where a common-mode range bias aliases most strongly, so a system required to obtain Z from ranging alone is the one that exposes the coupling. We name and stake the effect here; the companion paper (Paper B) supplies its identifiability analysis.

3 The Gap: A Narrow but Genuinely Unoccupied Combination

UWB self-calibration splits cleanly into two clusters (Table 1), echoing the two-cluster structure of the AutoPos concept's prior-art survey. The *inter-anchor-only* cluster is restricted to coplanar/2D deployments or requires a few manually surveyed reference anchors; the *3D* cluster achieves height observability only by adding an external sensor (LiDAR, SLAM, or visual-inertial odometry) or a mobile platform that traverses the space — effectively replacing manual surveying with a robot-assisted survey. The combination "3D + inter-anchor ranging only + joint antenna-delay + no known reference" is *unoccupied as a whole*, though every individual axis is occupied — we say so explicitly to avoid an axis-stacked "first." Joint anchor-plus-delay estimation already exists (Shah et al., 2022; Piavanini et al., 2022; De Preter et al., 2019); 3D inter-anchor-only self-calibration without a known reference exists in simulation (Pereira, 2021) and, via a *mobile* platform on hardware (Batstone et al., 2017). What is new is the *conjunction* on commodity static hardware with optical ground-truth validation, not any one ingredient. The conjunction is genuinely hard, and the field documented it as such and stepped *around* it rather than through it: De Preter et al. (2019) call the 3D calibration "very ill-conditioned" for similar tag/anchor heights and restrict to 2D with tape-measured heights; Corbalan et al. (2023) report vertical resolution "generally poor due to the low spread of anchors along that axis" and recommend measuring height independently; and commercial RTLS deployments that raise a few anchors to recover the vertical *survey* those raised positions rather than self-calibrate them. Each of these resolves the vertical by importing a measured height — the manual step AutoPos removes. AutoPos enters the regime deliberately and reports what actually happens when the vertical is left to ranging alone — which is where the companion paper's delay finding emerges.

Method	3D	Inter-anchor only	Delay est.	No known ref.	GT-validated
PANS auto-pos. (Decawave/Qorvo, 2019)	no	yes	no	yes	—
Ridolfi 2021 (Ridolfi et al., 2021)	no (2D)	yes	no	yes	2D
Piavanini 2022 (Piavanini et al., 2022)	no (2D)	yes	yes	yes	2D
Corbalan 2023 (Corbalan et al., 2023)	no (2D)	yes	no	**no**	yes (2D)
De Preter 2019 (De Preter et al., 2019)	no (2D)	**no** (mobile tag)	yes	**no**	RTK-GPS (2D)
Kok 2015 (Kok et al., 2015)	yes	**no**	clock only	**no**	yes
Shah 2022 (Shah et al., 2022)	yes	**no**	yes	**no**	yes
Pereira 2021 (Pereira, 2021)	yes	yes	no	yes	sim. only
Batstone 2017 (Batstone et al., 2017)	yes	**no** (mobile tag)	no	yes	real-hw
Nguyen 2025 (Nguyen et al., 2025)	yes	**no** (SLAM)	yes	yes	SLAM-dep.
Yuan 2024 (Yuan et al., 2024)	yes	**no** (LiDAR)	implicit	yes	LiDAR-dep.
Liu & Cao 2025 (Liu and Cao, 2025)	yes	**no** (LiDAR)	implicit	yes	LiDAR-dep.
Delama 2025 (Delama et al., 2025)	yes	**no** (VIO)	implicit	yes	VIO-dep.
AutoPos (this work)	**yes**	**yes**	**yes**	**yes**	**Vicon**

Table 1: Anchor self-calibration prior art. AutoPos occupies the otherwise-empty intersection of 3D, inter-anchor-ranging-only, joint antenna-delay estimation, and no known reference, validated against optical ground truth. "GT-validated" = evaluated against an independent ground-truth positioning system. "Implicit" delay = bias absorbed in optimization rather than explicitly separated.

4 What We Contribute

(1) The capability — and a reference point for later work. The headline result is a pure-UWB, self-calibrated vertical of ≈62 mm median absolute error (vs Vicon), near the ≈63 mm inter-anchor Cramér–Rao floor, from inter-anchor ranging alone — a reference point that later reference-free pure-UWB vertical self-calibration in this deployment regime can be measured against. It is produced by a complete, real-hardware pipeline (8× DWM1001C, two elevation levels) that takes bidirectional SS-TWR inter-anchor ranging to a 3D anchor layout with per-device antenna delays, using only commodity single-antenna TWR modules — two orders of magnitude cheaper than the AoA base stations (Zihajehzadeh and Park, 2017) or optical systems prior high-accuracy work relies on, and without the mobile sensor rigs the 3D self-calibration cluster requires. The deployed custom SS-TWR firmware applies a single generic antenna-delay default (the vendor example value, 16436 device units) identically to every node, with no per-device hardware calibration; the per-device delays the solver estimates are precisely the residuals this generic default leaves in the ranging. Calibration is thus pushed entirely into the AutoPos software solve — which is the realistic out-of-the-box deployment condition, and the physical reason a delay term is needed at all.

(2) The geometry that makes the cell enterable. A balanced two-layer 4+4 non-coplanar array. Non-coplanarity for 3D is itself known; our claim is narrower and is a statement about the *self-calibration* regime specifically: under reference-free inter-anchor self-calibration — and under the eight-anchor broadcast SS-TWR acquisition the array is dimensioned for — a *balanced* upper layer is what restores well-conditioned vertical *geometry*, whereas an asymmetric raise (e.g. one or two anchors lifted) leaves the vertical ill-conditioned. Deployed systems that raise a few anchors do not hit this wall — but only because they *survey* the raised positions rather than self-calibrate them, sidestepping the conditioning by importing the very measurement AutoPos removes. We quantify this with a self-calibration Fisher analysis (Figure 1): for a single coplanar layer the inter-anchor vertical Cramér–Rao bound is *exactly singular* ($\partial r/\partial z = 0$). A single raised anchor formally unlocks the vertical but leaves it badly conditioned (mean vertical CRLB ∼161 mm, worst-anchor ∼226 mm at 4+1); adding upper anchors improves it monotonically. Crucially the *mean* bound saturates early (∼81 mm at 4+2, ∼63 mm at 4+4), so an average alone would suggest stopping at two — but the *worst-conditioned* anchor keeps falling (226 → 134 → 122 → 93 mm across 4+1, 4+2, 4+3, 4+4). The balanced 4+4 is therefore the smallest array that brings *every* anchor's vertical to the production floor on average (∼63 mm, essentially the deployed median vertical error of ≈62 mm) and to no worse than ∼93 mm: the choice over an asymmetric 4+k raise is a worst-case-uniformity argument, not an observability one. At fixed anchor count a *balanced* (diagonal) pair beats a *clustered* (adjacent) pair by 1.24×, confirming that it is the balance, not merely the count, that conditions the vertical.

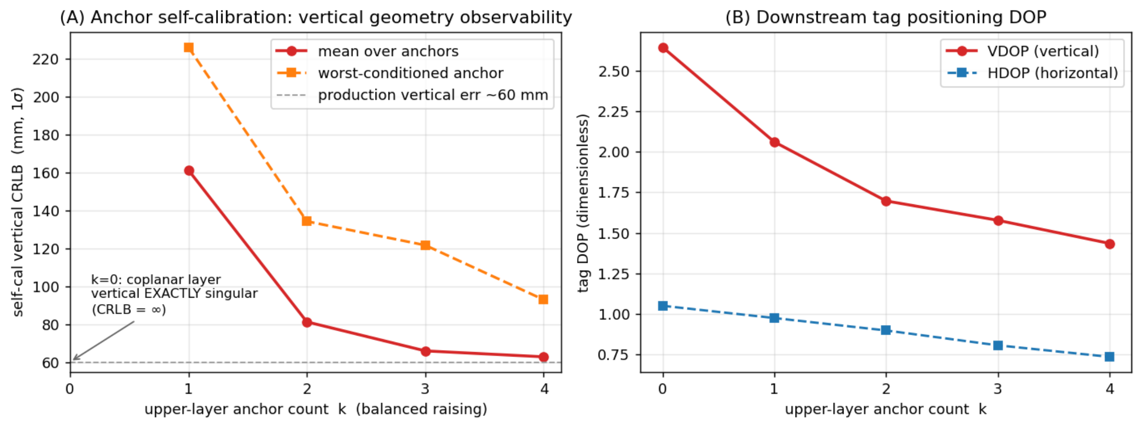

Figure 1: Vertical observability vs. upper-layer anchor count k (balanced raising). **(A)** Anchor self-calibration vertical CRLB from inter-anchor ranging: singular for a coplanar layer (k=0). The *mean* bound (circles) saturates near the ∼63 mm production vertical error by k=2, but the *worst-conditioned* anchor (squares) keeps improving to k=4 (226→134→122→93 mm), so the balanced 4+4 is the smallest array with *uniformly* well-conditioned vertical geometry — the case against an asymmetric 4+k raise. **(B)** Downstream tag-localization DOP for the same idealized balanced array: vertical (VDOP) improves monotonically with k but remains the worse-determined axis throughout (VDOP > HDOP). The deployment geometry's measured vertical DOP penalty is reported in the companion mechanism paper.

Why eight anchors, not more. The anchor count is set by a system constraint, not chosen for convenience — which answers the natural "why not 8+8 for better geometry?" To keep *moving* body-worn tags consistent, the ranging uses a **broadcast** scheme so that all anchors observe the same transmission instant, eliminating the motion-induced error that sequential per-anchor ranging accumulates as the tag moves; and the system is dimensioned for **10 Hz updates on up to 10 simultaneous tags**. That throughput and motion-coherence budget caps how many anchors can be ranged per cycle, so the balanced 4+4 is the *minimum* two-layer configuration that still reaches the vertical-observability floor (Figure 1) within the budget — a denser array such as 8+8 would either break the 10 Hz×10-tag throughput or reintroduce motion error. The claim is therefore not "use more anchors for better geometry" but "reach vertical observability with the fewest anchors a real-time multi-tag broadcast system can afford."

(3) Two deployment diagnostics that need no ground truth. Optical ground truth is exactly what is unavailable at deployment, so we identify two checks that flag a bad calibration using only the UWB data itself.

- *Ranging residual is a misleading quality signal.* With only four anchors — three position unknowns and one redundant measurement — the solver can fit the four distances almost perfectly even when the recovered position is wrong, so the configuration with the *lowest* residual can have the *worst* position. In our data the lowest-residual variant (11.5 mm) gives the worst positioning (∼169 mm), while the best positioner (49 mm) has a slightly *higher* residual (12.9 mm). The practical lesson: do not trust ranging residual as a calibration-quality score unless there are redundant anchors (≥ 6).
- *Dispersion across dynamic anchor selection exposes hidden bias.* When all 8 anchors are

used for every fix, any systematic calibration error appears as a constant offset and all solver variants look equally good (∼28–33 mm). The differences only appear when the tag selects a *different* 4-anchor subset per fix: a well-calibrated layout still yields consistent positions across subsets, whereas a biased layout yields scattered ones. The spread across subsets is therefore a calibration-quality indicator that a fixed anchor set would hide — and, crucially, it needs no ground truth.

(4) Honest validation. We convert the system from a repeatability claim to a ground-truth claim using optical motion capture, and report both the accuracy and the limit (next section).

5 What We Are *Not* Claiming (and the honest limits)

- **Not a new estimator.** SDP/MDS initialization, Tukey-bisquare IRLS, MAD/MVUE fusion, and alternating joint antenna-delay estimation are individually established. Joint estimation of anchor coordinates and per-anchor range bias from ranging is itself prior art (De Preter et al., 2019; Piavanini et al., 2022; Shah et al., 2022). Closest on the bias-estimation side is Yogesh et al. (2024), which estimates per-node additive range bias in a fully-connected UWB *swarm* validated against optical ground truth — but with the node positions *known* (from the optical system) and an explicitly pure-additive bias model, so it calibrates bias rather than self-calibrating geometry, and requires the very external reference AutoPos removes. The contribution here is the integration for the 3D inter-anchor-only, reference-free regime and the ground-truth characterization, not the algorithms.

- **The antenna delays are estimated and empirically useful, not yet independently validated as physical delays.** Their correctness is currently inferred only through downstream positioning and scale, so we report them as estimated-and-useful rather than "recovered." A factory-OTP cross-check was attempted but cannot serve this purpose: across all nine modules the DW1000 OTP stores only a near-uniform lot-level antenna-delay value with no per-device antenna calibration, so it does not supply the per-device delays the solver estimates — a finding that itself matters for the companion paper, where it is reported in full.

- **Consistency is not accuracy.** The preliminary headline numbers (positioning standard deviation of order 5 cm 3D / 4 cm vertical) measure repeatability of repeated fixes, not absolute error. Against optical ground truth the absolute error is larger, and only the ground-truth-validated figures are reported as accuracy.

- **The 4+4 geometry is necessary, not sufficient, for the vertical.** It completes the vertical *geometric* observability, but a systematic vertical bias remains after it — because that residual is not a geometry deficit but a *tag-side* delay aliasing into the vertical. The Fisher analysis makes this concrete: the inter-anchor vertical CRLB saturates near 63 mm for 4+4, at the level of the ≈62 mm deployed median vertical error — a bound on the *anchor* geometry's vertical contribution, not on the tag positioning error, but their closeness indicates the geometry is near its floor, so the residual vertical bias is consistent with a tag-side delay alias rather than a geometry deficit. This is the single most important honest caveat, and it is the tag-side face of the delay–layout coupling the companion paper analyzes, distinct from that paper's *headline* anchor-side delay↔scale result.

- **We do not claim general superiority of self-calibration over surveying** for applications demanding the highest achievable accuracy; we claim a favorable effort/accuracy/cost trade-off for rapid, equipment-free 3D deployment.

6 Accuracy by Axis, and the Pure-UWB Vertical

We report two distinct quantities and never conflate them: *repeatability* (standard deviation of repeated static fixes, the headline ∼5 cm figure) and *absolute accuracy* (error against Vicon, which is what the literature reports). The comparison to the landscape uses the absolute numbers.

Horizontal (X–Y plane). Repeatability ∼27 mm; absolute ∼37 mm (median vs Vicon). This sits in the same band as the horizontal accuracy of the best fusion systems (∼30–40 mm per axis). The horizontal is the easy axis for everyone; we neither lead nor lag here.

Vertical — the standout. Repeatability ∼41 mm; absolute ∼62 mm (median vs Vicon), from *pure UWB* with self-calibrated anchors, no IMU fusion, no AoA hardware, no surveyed anchors, and no external height sensor. This is the axis the field cannot get cheaply: 32 of the 37 reviewed systems avoid it (2D); the systems that do resolve it import a crutch a body-worn tag cannot use — AoA base stations plus biomechanical constraints plus OptiTrack (Zihajehzadeh and Park, 2017), or surveyed anchors plus IMU fusion. The pure-UWB vertical here lands in the band that others reach only with external help. **The defensible claim is not "smallest vertical number" but "a usable vertical recovered from UWB alone, under the body-worn constraint that rules out every external height aid"** — which is precisely why the balanced 4+4 array exists.

3D total. Repeatability ∼50 mm (∼28 mm when all eight anchors are visible online); absolute ∼73 mm median (P95 ≈172 mm, RMSE ≈110 mm). This sits in the upper range of the reviewed systems (10–80 cm) and within reach of the best (∼48 mm 3D RMSE), though we do *not* claim it is the single smallest absolute number: the best reviewed result (∼48 mm, a 24 s walk with surveyed anchors and IMU fusion) still edges it. The ≈73 mm median further benefits, in part, from the self-calibrated production layout's expanded scale compensating an uncorrected tag-side delay: with the tag-side delay left at zero, the clean metric-corrected layout gives a larger ≈110 mm median (the mechanism and its in-environment control are analyzed in the companion paper).

Honest caveat on the comparison. The reviewed systems are UWB+IMU *fusion* tracking a *moving* tag; AutoPos is UWB-only positioning of a *static* tag. Fusion and IMU help them; the static setting helps us. So the correct statement is that AutoPos reaches the same accuracy band *UWB-only and self-calibrated, with a resolved vertical*, not that it beats fusion systems.

Reproducibility scope — the acquisition scheme is part of the method. The positioning figures above are obtained at the system's designed operating point: an eight-anchor, two-layer **broadcast SS-TWR** acquisition in which every fix draws on the full anchor set. This is constitutive, not a favourable selection. A reproduction that drives the same layout with a naive minimal-anchor SS-TWR scheme, or that collapses the array to a single coplanar layer, will *not* recover these numbers — the vertical in particular returns to the ill-conditioned $\partial r/\partial z \to 0$ regime the Fisher analysis proves exactly singular for a coplanar layer. The graceful, layer-asymmetric degradation under partial anchor dropout (losing an *upper* versus a *lower* anchor is not equivalent) is characterised by the stratified keep-k analysis in the full system evaluation.

7 Relationship to the Companion Mechanism Paper

The companion mechanism paper (Paper B) is the natural home for the finding that this system surfaces: that the residual vertical error is the per-tag projection of a common-mode range bias — the UWB analogue of the GPS receiver-clock/altitude coupling — which no amount of added geometric diversity removes, only an explicit tag-delay correction (demonstrated there by an active vertical-injection control that produced no dose-response). Keeping the two separate avoids cannibalizing the higher-tier mechanism paper with the systems paper, and lets the mechanism paper cite this validated system as the substrate for its analysis. Paper A establishes priority on the deployment capability and the balanced 4+4 geometry; Paper B establishes priority on the delay–layout coupling — headlined by the anchor-side delay↔scale identifiability analysis and the falsification protocol, with the tag-side vertical aliasing above as the strand that bears on this system's residual limit.

8 Target Venue

A systems/deployment metrology venue: *IEEE Sensors*, *IEEE Access*, or *IPIN* (full paper). These reward a complete, ground-truth-validated, deployable system with a thorough ablation; the individually-known components are acceptable there given the integration and evaluation. The higher-tier *Measurement* / *IEEE TIM* target is reserved for the companion mechanism paper. Submitting Paper A first establishes the system as the cited substrate for Paper B.

9 Risk Assessment

- **"Incremental / just a combination."** Mitigated by leading with the empty-cell capability and the ground-truth-free diagnostics rather than the (obvious) two-layer idea, and by the honest validation. The hook is "first ground-truth-validated 3D inter-anchor-only self-calibration + deployment diagnostics," not "we used two layers."
- **"Does it actually work in 3D?"** Pre-empted by reporting the validated accuracy and the residual vertical limit openly, and by pointing to the companion mechanism paper rather than over-claiming.
- **"Two layers trivially fix Corbalan's Z problem."** Answered by the companion's finding: balanced 4+4 fixes the *geometric* part, but the vertical bias persists for a *delay* reason — so the contribution is not the layout but the demonstration that the obvious geometric fix is necessary-but-insufficient.
- **Prior-art overlap.** De Preter, Piavanini, Corbalan, Shah are cited prominently and delineated on the 3D / inter-anchor-only / no-known-reference axes; the overlap on *method* is acknowledged, and the contribution is scoped to capability and characterization.

10 Publication Strategy and Recommended Manuscript Front-Matter

Priority strategy. The most defensible novel element here is the balanced 4+4 array for *pure-UWB vertical observability* in reference-free self-calibration. To establish it as citable prior work, the recommended path is to keep the two papers separate and publish this system paper *first and early* (a public, dated, indexed record — e.g. an arXiv preprint followed by a fast systems venue), with the companion mechanism paper following and citing it. A merged paper would subordinate the 4+4 result to a delay-coupling narrative, reducing its discoverability and delaying the priority record. (The arXiv/early-disclosure decision interacts with institutional IP policy and patentability and should be agreed with the supervisor before posting.)

Enabling disclosure (for defensive-publication value). To maximise the blocking value of the priority record against a later overlapping claim, the manuscript states the full combination once, as a single enabling configuration: 3D anchor self-calibration; inter-anchor ranging only; joint per-device antenna-delay estimation; commodity single-antenna DWM1001C modules; a balanced two-layer 4+4 array; and optical-ground-truth validation — with no survey, no external Z sensor, and no mobile platform. A dated, indexed disclosure of this exact conjunction is what makes the combination both unpatentable by others and unambiguous to cite as prior art.

Reproducibility (a citation magnet, by design). The Fisher/CRLB derivation and figure ship as a runnable artifact, and the inter-anchor ranging dataset with solver outputs accompanies the preprint, so the ≈63 mm vertical floor and the by-axis accuracy can be reproduced independently. Releasing the derivation and data is part of the priority strategy: a reusable artifact that later work must cite to build on does more to lock in prior-art status than the claim sentence alone.

Recommended title (claim-forward, for discoverability): *"Pure-UWB Vertical Observability from a Balanced Two-Layer (4+4) Anchor Array: Reference-Free 3D Anchor Self-Calibration with Joint Antenna-Delay Estimation."* Alternatives retain the "AutoPos" name or lead with "Recovering the Vertical from UWB Alone."

Recommended abstract.

> Anchor-based ultra-wideband (UWB) positioning requires accurately known anchor coordinates, yet for body-worn motion capture the vertical axis has no shortcut: unlike a hovering platform, a body-worn tag cannot carry a downward range sensor or usefully fuse a barometer, so a drift-free, centimetre-level vertical reference can come only from the UWB geometry itself. We present a reference-free 3D anchor self-calibration that recovers anchor coordinates and per-device antenna delays from inter-anchor two-way ranging alone — no external sensor, no mobile survey platform, no surveyed reference positions — on commodity (DWM1001C) hardware, validated against optical (Vicon) ground truth. The enabling element is a balanced two-layer (4+4) anchor array: via a Fisher/Cramér–Rao analysis we show a single coplanar layer leaves the vertical exactly unobservable ($\partial r/\partial z = 0$), that raising anchors unlocks it, and that the balanced 4+4 array reaches a vertical CRLB floor of ≈63 mm, with *balance* — not merely anchor count — conditioning the vertical; the eight-anchor count itself follows from a real-time broadcast protocol dimensioned for 10 Hz on up to 10 simultaneous tags. Reporting by axis and separating repeatability from absolute accuracy, the system attains ≈5 cm 3D / ≈4 cm vertical repeatability and ≈7 cm 3D / ≈6 cm absolute (vs Vicon) — a pure-UWB vertical where most comparable systems avoid the axis or reach it only with angle-of-arrival, biomechanical, or surveyed-anchor aids. The 4+4 geometry is necessary but not sufficient: a residual systematic vertical bias remains, attributed to a tag-side delay aliasing into the vertical (companion paper). To our knowledge this is the first demonstration that the *vertical* axis of a UWB anchor array can be self-calibrated to the centimetre level — ≈62 mm median absolute against optical ground truth, near a ≈63 mm inter-anchor Cramér–Rao floor — from inter-anchor ranging alone on commodity single-antenna modules, with the delay coupling that bounds it identified and quantified.

Recommended keywords. ultra-wideband (UWB); anchor self-calibration; anchor auto-positioning; reference-free / infrastructure-free localization; inter-anchor two-way ranging (TWR); two-layer anchor array; vertical observability; 3D positioning; antenna-delay calibration; body-worn / wearable motion capture; Cramér–Rao bound; dilution of precision.

Citable priority sentence (to appear verbatim in the body): *"A balanced two-layer (*4+4*) anchor array renders the vertical* well-conditioned *for reference-free, inter-anchor-only 3D UWB self-calibration — a single coplanar layer is exactly singular (*$\partial r/\partial z = 0$*) and a single raised anchor only weakly unlocks it, while the balanced* 4+4 *reaches a* ≈63 *mm vertical Cramér–Rao floor — enabling a pure-UWB vertical without any external height aid."*

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